

FAIRMONT, MN, USA

WMO: 726586

Lat: 43.650N

Lon: 94.417W

Elev: 354

StdP: 97.15

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.6	-30.1	0.2	-23.0	-26.3	0.4	-20.7	15.4	-10.1	14.2	-11.7	4.7	330	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.2	23.5	30.8	22.8	28.8	21.6	25.2	29.8	24.0	28.4	23.0	27.4	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.8	19.4	28.4	22.5	18.0	27.2	21.4	16.7	26.0	79.0	29.7	74.1	28.5	69.7	27.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	11.9	10.8	-27.4	34.8	3.1	1.7	-29.6	36.0	-31.5	37.0	-33.2	38.0	-35.5	39.2		
			-27.5	27.1	3.0	1.2	-29.7	27.9	-31.4	28.6	-33.1	29.3	-35.3	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-8.9	-6.7	0.3	7.9	14.9	20.6	22.2	20.7	17.0	9.5	1.4	-5.9
	DBStd	12.27	7.20	7.08	6.84	5.48	4.84	3.56	3.04	3.03	4.56	5.15	6.16	6.60
	HDD10.0	2308	586	467	307	101	12	0	0	3	73	265	493	
	HDD18.3	4195	844	701	559	314	128	18	4	11	78	277	508	752
	CDD10.0	1513	0	0	7	38	164	317	379	331	213	57	7	0
	CDD18.3	360	0	0	0	2	23	84	126	84	38	3	0	0
	CDH23.3	3138	0	0	2	31	258	743	1082	626	359	36	1	0
	CDH26.7	934	0	0	0	7	83	240	339	158	101	7	0	0
Wind	WSAvg	5.1	5.8	5.8	5.7	6.0	5.7	4.7	3.7	3.4	4.2	5.0	5.6	5.6
Precipitation	PrecAvg	831	20	21	49	84	106	118	110	106	86	59	44	26
	PrecMax	1330	59	80	125	160	230	369	293	386	258	202	144	109
	PrecMin	497	1	0	6	8	32	38	20	18	15	2	2	2
	PrecStd	159	16	16	28	41	51	61	59	71	58	48	36	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	12.3	21.4	27.4	32.5	33.8	33.0	32.5	32.2	27.3	21.0	12.3
		MCWB	4.1	8.7	13.1	16.1	20.4	22.4	25.4	24.7	21.8	18.5	12.7	9.2
	2%	DB	3.8	7.5	17.1	22.8	28.8	31.4	32.2	30.1	29.1	23.0	17.0	7.5
		MCWB	1.7	5.0	11.9	13.8	17.9	22.0	24.9	23.8	20.9	15.7	11.3	4.5
	5%	DB	2.3	4.3	12.8	19.9	26.3	29.1	30.1	28.0	27.4	21.0	12.9	4.0
		MCWB	0.8	2.4	9.1	12.2	16.4	20.6	23.7	22.4	19.4	14.5	8.7	1.9
	10%	DB	1.0	2.5	9.1	17.2	23.0	27.5	28.0	27.1	25.2	17.7	10.1	2.4
		MCWB	-0.2	0.9	6.5	10.8	15.0	19.4	22.3	21.4	18.9	12.4	6.8	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	9.3	15.6	17.4	22.1	25.1	27.5	26.4	23.8	21.4	14.4	9.8
		MCDB	7.6	11.9	18.6	24.2	29.2	31.1	31.7	30.8	29.1	24.7	19.0	12.3
	2%	WB	2.0	5.1	12.5	15.1	19.9	23.6	25.6	24.7	22.4	17.3	11.7	4.7
		MCDB	3.6	7.4	16.0	20.8	26.0	29.0	30.1	28.7	27.4	21.2	15.9	7.6
	5%	WB	0.8	2.8	9.7	13.1	18.1	22.2	24.5	23.5	21.0	15.2	9.1	2.4
		MCDB	2.0	4.6	12.9	18.6	23.7	27.7	28.8	27.3	25.6	19.4	12.7	4.2
	10%	WB	-0.4	1.0	6.8	11.4	16.5	20.8	23.5	22.2	19.7	12.9	6.9	0.8
		MCDB	0.6	2.5	9.5	16.1	21.6	25.8	27.7	25.8	23.7	17.5	9.8	2.2
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	8.8	10.7	11.0	10.5	10.4	10.3	12.0	10.9	9.2	8.0
		MCDBR	9.6	10.0	13.9	16.6	15.1	13.0	11.7	11.4	14.1	15.1	13.9	10.8
		MCWBR	8.0	8.2	9.5	9.4	7.2	6.2	6.5	6.7	7.1	8.7	9.1	8.5
	5% WB	MCDBR	8.8	9.5	12.8	13.8	12.4	11.1	10.5	10.3	11.9	12.6	12.8	10.2
		MCWBR	7.6	7.8	9.2	8.9	7.0	6.5	6.6	6.7	6.9	8.3	9.0	8.3
Clear-Sky Solar Irradiance	taub	0.271	0.285	0.313	0.365	0.384	0.406	0.417	0.409	0.367	0.327	0.286	0.264	
	taud	2.437	2.408	2.440	2.328	2.345	2.332	2.281	2.290	2.376	2.486	2.523	2.479	
	Ebn at Noon	873	918	930	899	888	866	850	845	857	856	849	850	
	Edh at Noon	72	89	99	120	122	124	129	124	105	83	67	63	
All-Sky Solar Radiation	RadAvg	1.83	2.88	3.79	4.67	5.47	6.06	6.48	5.39	4.33	2.86	1.90	1.39	
	RadStd	0.20	0.33	0.39	0.45	0.39	0.47	0.27	0.39	0.29	0.37	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page